

Chapter 7 Analysis and Design of Linear Discrete-Time System (Sampled-data System)

7.1 Introduction

7.2 The Sampling Process and Sampling Theorem

7.3 Signal Recovery and Zero-Order Hold

7.4 Z-Transform and Inverse Z Transform

7.5 Mathematical Models of Discrete-Time Systems

7.6 Performance Analysis of Discrete-Time Systems

7.7 Digital Control Design for Discrete-Time Systems

Design for discrete-time systems can be done in s-domain, z-domain and w-domain, respectively.

Design for discrete-time systems:

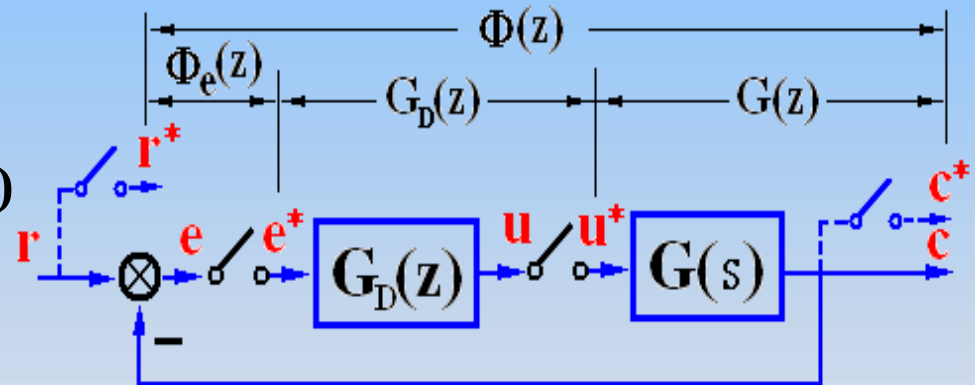
{ Design by discrete equivalent (Emulation) $\Rightarrow$ Continuous compensation
 { Discrete Design: Digital compensation

7.7.1 The Impulse Transfer Function for the Digital Controller

$$\Phi(z) = \frac{G_D(z) \cdot G(z)}{1 + G_D(z) \cdot G(z)}$$

$$\Phi_e(z) = \frac{1}{1 + G_D(z) \cdot G(z)} = 1 - \Phi(z)$$

$$G_D(z) \cdot G(z) = \frac{\Phi(z)}{1 - \Phi(z)} = \frac{\Phi(z)}{\Phi_e(z)}$$



$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)}$$

$$\Phi_e(z) = 1 - \Phi(z), E(z) = \Phi_e(z)R(z)$$

7.7.2 Deadbeat Control Design 最少拍控制

Deadbeat Control Systems: Matching a particular test input within a number of steps. — No steady-state error on the sampling point.

(典型输入作用下，能在有限拍内结束响应过程且在采样点上无稳态误差的系统。)

1. A unified description of typical test inputs

$$r(t) = \begin{cases} 1(t) \\ t \\ t^2/2 \end{cases} \quad R(z) = \begin{cases} \frac{z}{z-1} = \frac{1}{1-z^{-1}} \\ \frac{Tz}{(z-1)^2} = \frac{Tz^{-1}}{(1-z^{-1})^2} \\ \frac{T^2 z(z+1)}{2(z-1)^3} = \frac{T^2 z^{-1}(1+z^{-1})}{2(1-z^{-1})^3} \end{cases} \quad \frac{A(z)}{(1-z^{-1})^\nu}$$

ν	$A(z)$
1	1
2	Tz^{-1}
3	$\frac{T^2 z^{-1}(1+z^{-1})}{2}$

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)}$$

Design Idea: Obtain $G_D(z)$ by constructing $\Phi(z)$ so that the output can match the typical test signal within the minimum steps.

No $\begin{cases} \text{Zeros} \\ \text{Poles} \end{cases}$ of $G(z)$ on or beyond the unit circle, except for $(1, j0)$

$$R(z) = \frac{A(z)}{(1 - z^{-1})^\nu}$$

$$E(z) = \Phi_e(z)R(z), \quad \Phi_e(z) = 1 - \Phi(z)$$

$$e(\infty) = \lim_{z \rightarrow 1} (z - 1) \Phi_e(z) R(z) \implies \Phi_e(z) = (1 - z^{-1})^\nu F(z^{-1})$$

To make the $G_D(z)$ simplest and of the lowest-order, we can choose $F(z^{-1})$ as 1.

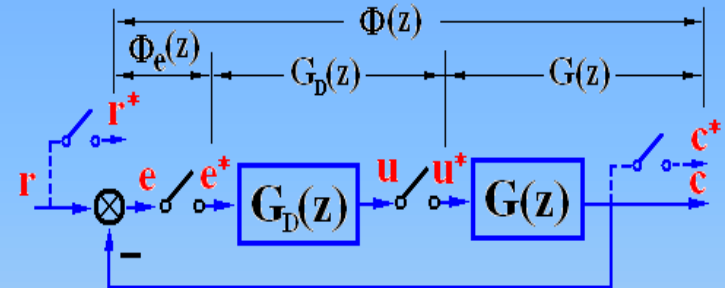
$$\Phi(z) = 1 - \Phi_e(z) = 1 - (1 - z^{-1})^\nu$$

From the design idea, we know that $e(\infty T) = 0$

$$E(z) = \Phi_e(z) \cdot R(z) = \frac{A(z)}{(1-z^{-1})^\nu} \Phi_e(z)$$

$$e(\infty T) = \lim_{z \rightarrow 1} (1-z^{-1}) \frac{A(z)}{(1-z^{-1})^\nu} \Phi_e(z) = 0$$

$$\Phi_e(z) = (1-z^{-1})^\nu F(z) \stackrel{F(z)=1}{=} (1-z^{-1})^\nu$$



Hence:

$$\begin{aligned} \Phi(z) &= 1 - \Phi_e(z) = 1 - (1-z^{-1})^\nu = b_1 z^{-1} + b_2 z^{-2} + \dots + b_\nu z^{-\nu} \\ &= \frac{b_1 z^{\nu-1} + b_2 z^{\nu-2} + \dots + b_\nu}{z^\nu} \end{aligned}$$

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)}$$

The rule to construct $\Phi(z)$: All poles of $\Phi(z)$ are located on the origin of z-plane.

2. $\Phi(z)$ for typical test inputs

(1) for $r(t) = 1(t)$

- The C.L.impulse transfer function:

$$v = 1 \quad \Phi(z) = z^{-1}$$

$$E(z) = 1$$

The system can track the input by 1 step only.

(2) for $r(t) = t \cdot 1(t)$

- **The C.L.impulse transfer function:**

$$\nu = 2 \quad \Phi(z) = 2z^{-1} - z^{-2}$$

$$E(z) = Tz^{-1}$$

The system can track the input by 2 step.

(3) for $r(t) = \frac{1}{2}t^2 \cdot 1(t)$

- The C.L.impulse transfer function:

$$\nu = 3 \quad \Phi(z) = 3z^{-1} - 3z^{-2} + z^{-3}$$

$$E(z) = \frac{1}{2}T^2 z^{-1} + \frac{1}{2}T^2 z^{-2}$$

The system can track the input by 3 step.

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)}$$

Deadbeat Control Design Table

$r(t)$	$R(z)$	$\Phi_e(z) = (1 - z^{-1})^v$	$\Phi(z) = 1 - \Phi_e(z)$	$G_D(z)$	t_s
$1(t)$	$\frac{1}{1 - z^{-1}}$	$1 - z^{-1}$	z^{-1}	$\frac{z^{-1}}{(1 - z^{-1}) \cdot G(z)}$	T
t	$\frac{Tz^{-1}}{(1 - z^{-1})^2}$	$(1 - z^{-1})^2$	$2z^{-1} - z^{-2}$	$\frac{z^{-1}(2 - z^{-1})}{(1 - z^{-1})^2 G(z)}$	$2T$
$\frac{t^2}{2}$	$\frac{T^2 z^{-1}(1 + z^{-1})}{2(1 - z^{-1})^3}$	$(1 - z^{-1})^3$	$3z^{-1} - 3z^{-2} + z^{-3}$	$\frac{z^{-1}(3 - 3z^{-1} + z^{-2})}{(1 - z^{-1})^3 G(z)}$	$3T$

3. Algorithm for Deadbeat Control Design

① Obtain $G(z)$ — Suppose there are no poles and zeros of $G(z)$ on or beyond the unit circle.

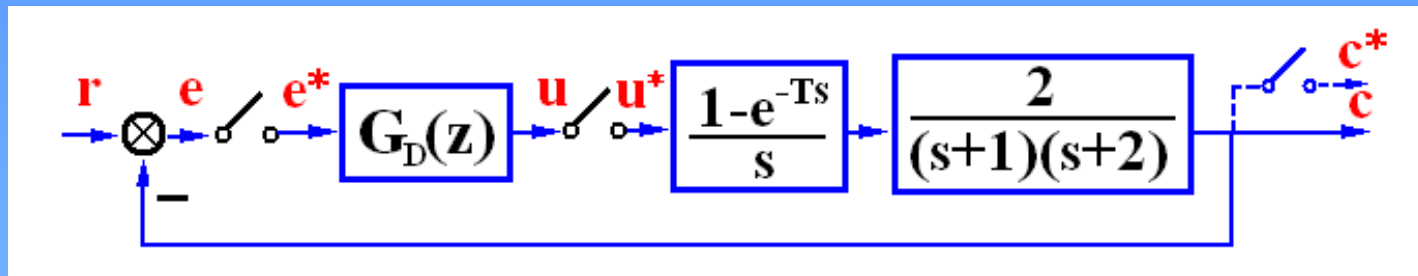
② Determine $\Phi_e(z)$ for the particular test input

$$r(t) \Rightarrow R(z) = \frac{A(z)}{(1-z^{-1})^\nu} \Rightarrow \Phi_e(z) = (1-z^{-1})^\nu$$

$$\Phi_e(z) = (1-z^{-1})^\nu F(z)$$

③ Obtain $\Phi(z) = 1 - \Phi_e(z)$

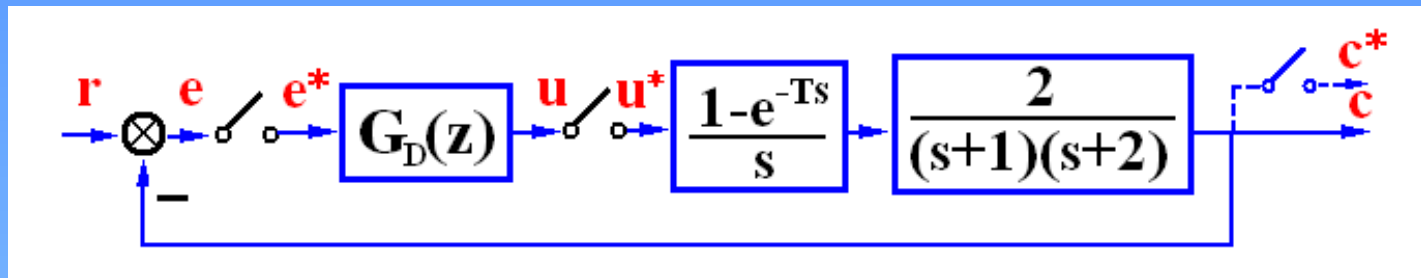
④ Achieve $G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)}$



Example 1. Consider the system shown in the above figure ($T=1$). Design deadbeat controllers $G_D(z)$ for $r(t)=1(t)$ 、 t .

Solution.

$$\begin{aligned}
 G(z) &= Z \left[\frac{1-e^{-Ts}}{s} \cdot \frac{2}{(s+1)(s+2)} \right] = 2(1-z^{-1}) \cdot Z \left[\frac{C_0}{s} - \frac{C_1}{s+1} + \frac{C_2}{s+2} \right] \\
 &= 2 \cdot \frac{z-1}{z} \cdot Z \left[\frac{1}{2} \cdot \frac{1}{s} - \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{s+2} \right] \\
 &= \frac{z-1}{z} \left[\frac{z}{z-1} - \frac{2z}{z-e^{-T}} + \frac{z}{z-e^{-2T}} \right] = 1 - \frac{2(z-1)}{z-e^{-T}} + \frac{z-1}{z-e^{-2T}} \\
 &= \frac{(1+e^{-2T}-2e^{-T})z + (e^{-3T}+e^{-T}-2e^{-2T})}{(z-e^{-T})(z-e^{-2T})} \\
 &\stackrel{T=1}{=} \frac{0.4(z+0.365)}{(z-0.368)(z-0.136)}
 \end{aligned}$$



Referring to the result for $r(t) = 1(t)$ in the Design Table

$$R(z) = \frac{z}{z-1} \quad \text{Choose} \quad \begin{cases} \Phi_e(z) = 1 - z^{-1} \\ \Phi(z) = 1 - \Phi_e(z) = z^{-1} \end{cases}$$

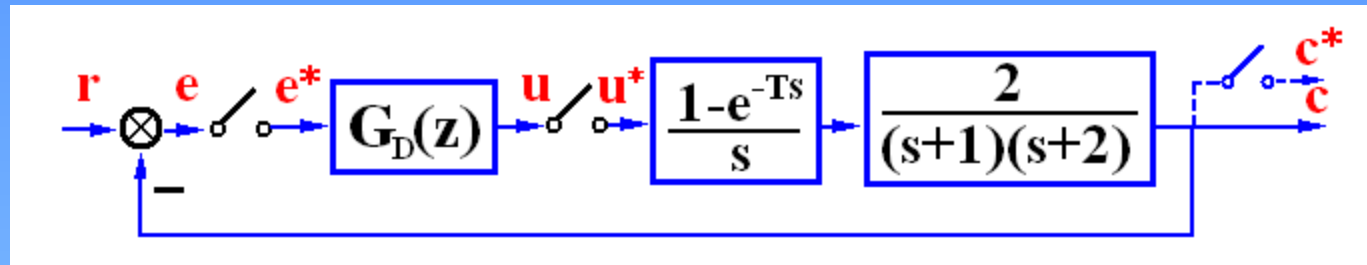
$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)} = \frac{z^{-1}}{1 - z^{-1}} \cdot \frac{(z - 0.368)(z - 0.136)}{0.4(z + 0.365)}$$

$$= \frac{2.5(z - 0.368)(z - 0.136)}{(z - 1)(z + 0.365)}$$

$$C(z) = \Phi(z)R(z) = z^{-1} \cdot \frac{1}{1 - z^{-1}}$$

$$= z^{-1} [1 + z^{-1} + z^{-2} + \dots] = z^{-1} + z^{-2} + z^{-3} + \dots$$

$$E(z) = \Phi_e(z)R(z) = (1 - z^{-1}) \cdot \frac{1}{1 - z^{-1}} = 1$$



For $r(t) = t$

$$R(z) = \frac{Tz^{-1}}{(1 - z^{-1})^2} \quad \text{Choose} \quad \begin{cases} \Phi_e(z) = (1 - z^{-1})^2 \\ \Phi(z) = 1 - \Phi_e(z) = 2z^{-1} - z^{-2} \end{cases}$$

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z) \cdot G(z)} = \frac{2z^{-1} - z^{-2}}{(1 - z^{-1})^2} \cdot \frac{(z - 0.368)(z - 0.136)}{0.4(z + 0.365)}$$

$$= \frac{5(z - 0.5)(z - 0.368)(z - 0.136)}{(z - 1)^2(z + 0.365)}$$

$$E(z) = \Phi_e(z) \cdot R(z) = Tz^{-1}$$

$$C(z) = \Phi(z)R(z) = (2z^{-1} - z^{-2}) \cdot \frac{Tz^{-1}}{(1 - z^{-1})^2}$$

$$= R(z) - E(z) = 2Tz^{-2} + 3Tz^{-3} + 4Tz^{-4} + \dots$$

Although the deadbeat control system tracks a particular test input accurately within a number of steps, it has the following disadvantages:

- (1) It is designed only for a particular input.**
- (2) The output has ripples although there are no errors on the sampling points.**
- (3) The control input changes drastically.**

4. $G(z)$ has poles or zeros on or beyond the unit circle
suppose

$$G(z) = \frac{z^{-v} \prod_{i=1}^L (1 - z_i z^{-1})}{\prod_{i=1}^n (1 - p_i z^{-1})}$$

**where z_i is the zero of $G(z)$;
 p_i is the pole of $G(z)$.**

Then

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z)G(z)} = \frac{z^v \prod_{i=1}^n (1 - p_i z^{-1}) \Phi(z)}{\prod_{i=1}^L (1 - z_i z^{-1}) \Phi_e(z)}$$

$$G_D(z) = \frac{\Phi(z)}{\Phi_e(z)G(z)} = \frac{z^v \prod_{i=1}^n (1 - p_i z^{-1}) \Phi(z)}{\prod_{i=1}^L (1 - z_i z^{-1}) \Phi_e(z)}$$

① If there is z^v in $G_D(z)$, $G_D(z)$ is un-realizable.

Thus, we have to ensure that there exists z^{-v} in $\Phi(z)$, which promises $G_D(z)$ is realizable.

② If there is z_i on or beyond the unit circle, $G_D(z)$ is unstable.

Then, those z_i will be designed as the zeros of $\Phi(z)$.

③Note that

$$\Phi(z) = G_D(z)G(z)\Phi_e(z)$$

If there are p_i on or beyond the unit circle,

$\Phi(z)$ will be unstable,

Then those p_i will be designed as the zeros of $\Phi_e(z)$.

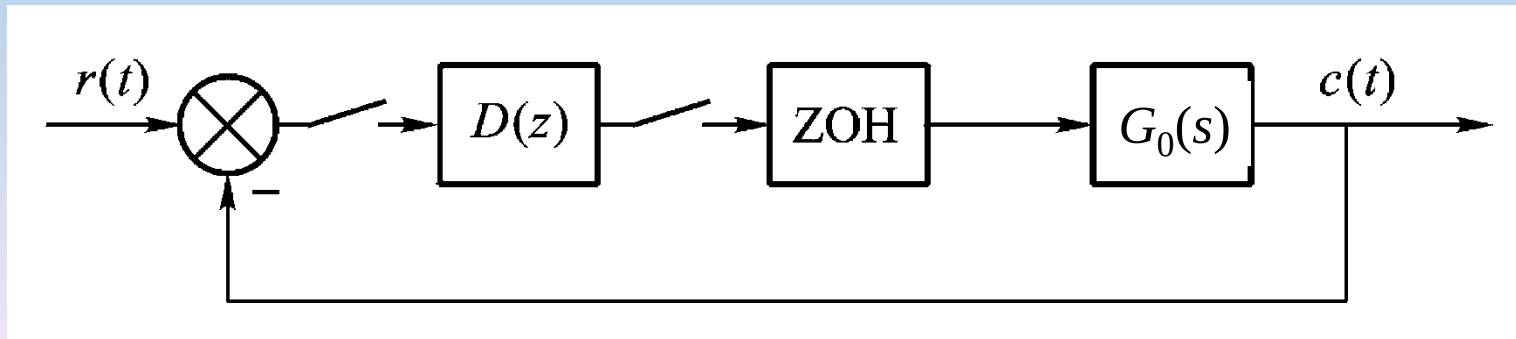
Example Given the discrete system described as in the following figure, where

$$G_0(s) = \frac{10}{s(0.1s + 1)(0.05s + 1)}, \quad G_h(s) = \frac{1 - e^{-Ts}}{s}$$

with

$$T = 0.2s$$

Design a deadbeat controller for $r(t) = 1(t)$



Solution: the O. L. impulse transfer function is

$$G(z) = Z[G_h(z)G_0(z)] = \frac{0.76z^{-1}(1+0.05z^{-1})(1+1.065z^{-1})}{(1-z^{-1})(1-0.135z^{-1})(1-0.0185z^{-1})}$$

For $r(t) = 1(t)$, we can design

$$\Phi_e(z) = 1 - z^{-1} \quad (1)$$

$$\Phi(z) = z^{-1} \quad (2)$$

Because there exists $z = -1.065$ (beyond the unit circle),

Thus, z should also be the zero of $\Phi(z)$

There exist z^{-1} in $G(z)$, z^{-1} should be in $\Phi(z)$, thus

$$\Phi(z) = z^{-1}(1+1.065z^{-1}) \quad (3)$$

Because that

$$\Phi(z) = 1 - \Phi_e(z) \quad (4)$$

**from (3), $\Phi(z)$ is now a polynomial on z^{-1} of order 2,
To satisfy (4) , $\Phi_e(z)$ must be a polynomial on z^{-1}
of order 2, thus based on (1), we redesign:**

$$\Phi_e(z) = (1 - z^{-1}) (1 + a_1 z^{-1}) \quad (5)$$

Where a_1 is a constant to be chosen later.

Thus multiplied by a constant b_1 to be designed later, we get

$$\Phi(z) = b_1 z^{-1} (1 + 1.065 z^{-1}) \quad (6)$$

From (5) and (6) , we get:

$$a_1 = 0.516 \quad b_1 = 0.484$$

Thus,

$$\Phi_e(z) = (1 - z^{-1}) (1 + 0.516z^{-1}) \quad (7)$$

$$\Phi(z) = 0.484z^{-1}(1 + 1.065z^{-1}) \quad (8)$$

Then the deadbeat controller is

$$\begin{aligned} D(z) &= \frac{1 - \Phi_e(z)}{G(z)\Phi_e(z)} \\ &= \frac{1 - (1 - z^{-1}) (1 + 0.516z^{-1})}{\frac{0.76z^{-1}(1 + 0.05z^{-1}) (1 + 0.065z^{-1})}{(1 - z^{-1}) (1 - 0.135z^{-1}) (1 - 0.0185z^{-1})} (1 - z^{-1}) (1 + 0.516z^{-1})} \end{aligned}$$

$$D(z) = \frac{0.637(1 - 0.0185z^{-1})(1 - 0.135z^{-1})}{(1 + 0.05z^{-1})(1 + 0.516z^{-1})}$$

Then the Z-transform is

$$\begin{aligned} C(z) &= \Phi(z)R(z) = 0.484z^{-1}(1 + 1.085z^{-1})\frac{1}{1 - z^{-1}} \\ &= 0.484z^{-1} + z^{-2} + z^{-3} + \dots + z^{-4} + \dots \end{aligned}$$

System can follow the input at the 2nd step, which is one step later.

5. Ripple-free deadbeat control design

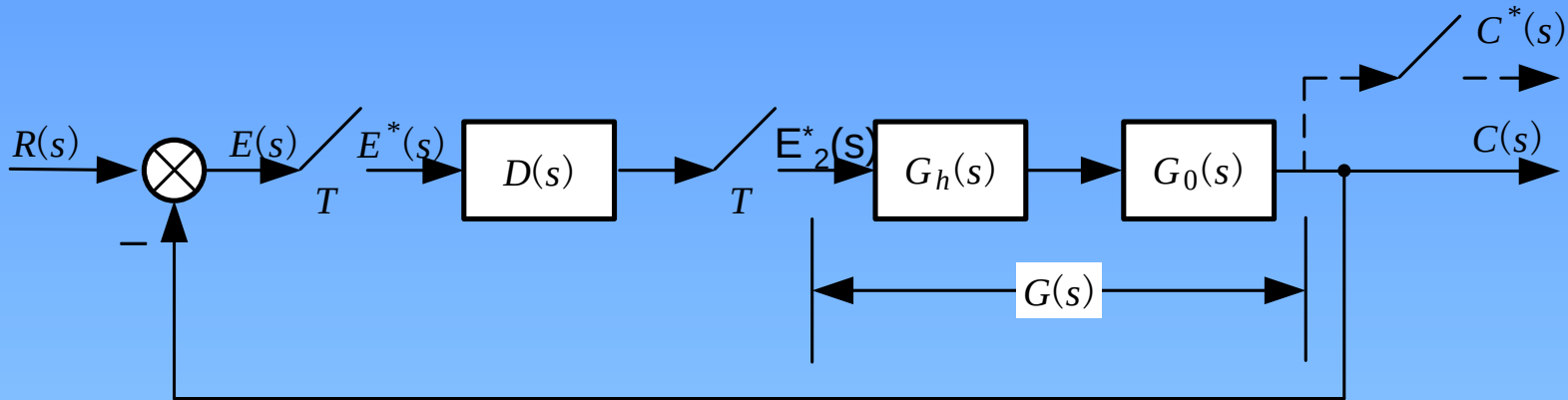
Ripple: though the system outputs are stable at the sampling time, they are varying between two sampling time, 见p251,图7-50。

Objective: Not only tracking the input at the sampling time, the outputs are ripple-free.

Necessary Condition: Given a input signal

$$r(t) = R_0 + R_1 t + \frac{1}{2} R_2 t^2 + \cdots + \frac{1}{(q-1)!} R_{q-1} t^{q-1}$$

Then $G_0(s)$ exists $(q-1) z$.



$$E_2(z) = D(z)E(z)$$

solution: ensure $E_2(z)$ being a polynomial on z^{-1} of a finite order.

Condition: $E_2(z)$ is a polynomial on z^{-1} of finite order.

$$E_2(z) = D(z)E(z) = D(z)\Phi_e(z)R(z), \quad D(z)\Phi_e(z) = \frac{\Phi(z)}{G(z)}$$

→ the zero of $G(z)$ must be a zero of $\Phi(z)$

最少拍设计中， $\Phi(z)$ 和 $\Phi_e(z)$ 选取时应遵循的原则：

1. $G_D(z)$ 零点的数目不能大于极点的数目；
2. $\Phi_e(z)$ 应把 $G(z)$ 在单位圆上及单位圆外的极点作为自己的零点；
3. $\Phi(z)$ 应把 $G(z)$ 在单位圆上及单位圆外的零点作为自己的零点；
4. 当 $G(z)$ 含有 z^{-1} 因子时，要求 $\Phi(z)$ 也含有 z^{-1} 的因子；
5. 因为 $\Phi(z)=1-\Phi_e(z)$ ，他们应该是关于 z^{-1} 同样阶次的多项式，而且 $\Phi_e(z)$ 还应包含常数项1。
6. 当最小拍系统还有无纹波要求时，闭环脉冲传函 $\Phi(z)$ 的零点应抵消 $G(z)$ 的全部零点（因为最少拍系统设计中 $G(z)$ 单位圆上及单位圆外的零极点已经被补偿，因此在无纹波的设计中只需抵消 $G(z)$ 单位圆内的零点）。